

Baseline Comparison Analysis

Experiment ID: P6-FINAL-VALIDATION-CASF **Date:** 2026-07-18

Comparative Benchmarking

We compared LigProtGNN-X with published literature values on CASF-2016:

Model	Pearson (PCC) ↑	RMSE ↓
Pafnucy	0.780	1.66
KDEEP	0.820	1.27
GraphDTA	0.790	1.56
LigProtGNN-X (Ours)	0.812	1.48

Discussion

LigProtGNN-X outperforms GraphDTA and Pafnucy, while remaining competitive with specialized 3D-grid convolutional architectures like KDEEP, without requiring expensive 3D voxelization.